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# High-performance fire detection framework based on feature enhancement and multimodal fusion

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ABSTRACT

Fire detection technology has become increasingly critical in the context of rising global fire threats and extreme weather conditions. Traditional methods rely on single-modal sensors and conventional image processing techniques and often struggle with complex environmental variations and background interference. This study proposes an innovative multimodal fire detection framework that integrates advanced deep learning techniques to address these limitations. By leveraging a comprehensive approach that combines YOLOv8-based object detection, HSV color space enhancement, completed local binary pattern (CLBP) texture analysis, and a novel dynamic feature enhancement module (DFEM), the proposed method significantly improves fire detection accuracy and robustness. This research introduces a sophisticated multimodal fusion strategy that systematically processes fire-related features across multiple domains. A key innovation is the cross-modality fusion Mamba (CMFM) module, which employs efficient channel attention (ECA) and an efficient 2D-selective scan module (E2DSM) to dynamically integrate and refine features from different modalities. Experimental validation was conducted on a dataset that we collected, which was supplemented by data collected via real-world robotic image acquisition in diverse environments, including forests, corridors, and outdoor settings. The proposed method demonstrated exceptional performance, with a precision of 96.4%, a recall of 95.7%, and an overall accuracy of 95.8%, outperforming state-of-the-art models such as VGG16, ResNet50, YOLOv5, and YOLOv8. Ablation studies further validated the contribution of each module and highlighted the framework's robust feature enhancement and fusion capabilities.

## 1. Introduction

With the continuous advancement of globalization and urbanization, various environments (such as high-rise buildings, dense communities and natural areas) are facing increasingly severe threats from fires. Against the backdrop of global climate change and frequent extreme weather, the frequency and intensity of fires continue to rise, and the resulting economic losses, ecological damage and casualties have increased significantly. In recent years, large-scale fires in California and Australia have further highlighted the severity of this problem. In this context, timely and accurate fire detection technology has become an indispensable part of fire prevention and control and emergency response systems. Timely warning at the early stage of a fire can not only minimize losses but also provide valuable time for emergency rescue [1].

Traditional methods for fire detection, such as those based on smoke, temperature, and flames, have long served as primary mechanisms for fire alarms. These systems typically rely on sensors installed at specific locations to conduct real-time sampling of smoke particle density or sudden temperature changes. However, these methods are highly sensitive to environmental factors [2]. With technological advancements, visual image processing technology has provided an alternative avenue for fire detection. This method does not depend solely on readings from physical sensors. Instead, it uses cameras to capture images and further discerns the presence of fire by analyzing color, texture, and shape information related to flame characteristics. This integration of visual information offers greater flexibility and a wider range of application scenarios for fire detection. Related research has also gradually been carried out. Celik et al. [3] introduced a generic rule-based color model for the detection of fire pixels. Wang et al. [4] proposed a flame detection method based on area thresholds.

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Maruta et al. [5] developed a smoke detection technique that uses image texture features combined with support vector machines. Foggia et al. [6] suggested an approach for fire detection that incorporates supplementary information on color, shape variation, and motion analysis. However, features extracted by traditional methods can encounter challenges in certain scenarios. Specifically, when the background of a scene contains colors and textures similar to flames or when the lighting conditions are poor, such as during sunset or in the presence of strong light reflections, conventional image processing techniques may misidentify these regions as fires, thereby leading to false alarms. Consequently, although the above methods have detected fires successfully in numerous instances, the reliability of these techniques remains a critical issue that needs to be addressed through further research and improvement in the face of constantly changing environments and fire characteristics. Moreover, these methods often do not incorporate infrared information, which is crucial for detecting fire-related temperature variations. This limitation is addressed by fusing visible light and infrared images, thereby combining their complementary strengths to improve detection accuracy and robustness.

As deep learning has made strides in the field of computer vision, its application in fire detection research has garnered widespread attention. By training on large datasets, deep learning models can autonomously learn and extract complex features related to fires, which increases their adaptability to complex scenarios and environmental factors. In recent years, the domain of fire detection research has expanded from convolutional neural networks (CNNs) to more complex and powerful deep learning architectures [7]. Advanced object detection networks such as Faster R-CNN [8], EfficientNet [9], and the YOLO series of models [10] have achieved remarkable results in fire detection [11–13]. However, these detection models rely mainly on single-modal images as network inputs, and this single data source approach is often susceptible to interference from environmental factors. Under certain lighting or complex background conditions, the models may misinterpret the captured image features. For example, some objects may have characteristics similar to those of fires (color and shape) or may resemble fires due to factors such as environmental changes (such as clouds, fog, and light reflection). Single-modal images may not fully reflect the effective characteristics of flames, thereby affecting the accuracy of fire detection.

Multimodal fusion compensates for the shortcomings of single-modal methods by integrating information from multiple data sources [14]. In this study, "multimodal" refers to data obtained via different imaging techniques or sensors, even if they share the same format. Specifically, we treat visible light images and their texture feature images as distinct modalities, as they capture different types of information — visual content and texture patterns — and are complementary. This broader interpretation is consistent with common practices in related works, where different aspects of the same imaging modality are treated as independent sources of information. Significant progress has been made in the fields of medical treatment [15] and remote sensing [16]. On the basis of multimodal fusion, the use of ResNet as a multi-input network structure can further improve the performance of fire detection. In contrast to the traditional CNN architecture, ResNet successfully solves the gradient vanishing/exploding problem in deep networks, making the training of very deep networks possible. Since its introduction in 2015, ResNet has been the gold standard architecture in the field of computer vision and has achieved leading performance in many computer vision tasks. Studies have shown that EfficientNetB7 outperforms ResNet50 in terms of classification accuracy [9]. In addition, transformer-based models [17] perform competitively in various tasks. However, classic ResNet architectures still demonstrate significant competitiveness compared with transformer-based models in large-scale image recognition tasks [18]. Although ResNet has achieved great success in various computer vision tasks, its application in multimodal data, especially when integrating different types of image features, still faces various challenges. For example, when fusing information from different modalities such as color and texture, the network has difficulty capturing fine-grained details such as texture features. These texture details are crucial for accurate classification, but they are easily lost during the feature fusion process, which results in poor performance. In addition, while existing fusion methods attempt to combine modality-specific features, they are usually unable to adjust to the differences between modalities dynamically, which can lead to incorrect representations of key features.

In summary, the existing methods have the following shortcomings:

1. Traditional fire detection methods based on color and texture often fail to identify flames accurately in dynamic settings, where background elements can resemble fire features, thereby leading to poor performance in real-world applications.

2. Deep learning models, including CNNs and YOLO, use primarily single-modal image inputs, which makes them vulnerable to issues such as lighting changes, reflections, and clutter. This limits their ability to detect fires reliably under varied and challenging conditions.

3. Although multimodal fusion can improve detection, current methods often fail to effectively combine features such as color and texture, which results in the loss of critical information. Additionally, these fusion techniques do not dynamically adjust to modality differences, which limits their adaptability to real-world conditions.

Given the aforementioned limitations of both traditional fire detection methods and single-modal deep learning approaches, this paper proposes a more comprehensive and robust multimodal fire detection framework. Building upon the strengths of existing object detection architectures while addressing their shortcomings, our method integrates object detection, color and texture feature analysis, and advanced feature fusion techniques to better adapt to complex environments and dynamic fire characteristics. The innovations of this framework are as follows:

1. This paper employs a YOLOv8-based preprocessing stage to identify potential fire regions efficiently, which ensures high precision and speed even in challenging environments with cluttered backgrounds and varying lighting conditions. The adaptability of YOLOv8 enables the model to focus on relevant areas and significantly improves detection accuracy.

2. By combining HSV color space enhancement and completed local binary pattern (CLBP) texture analysis, the method captures complementary features that improve detection performance. This fusion of color and texture information enhances the ability of the model to detect subtle fire characteristics under complex environmental conditions.

3. A dynamic feature enhancement module (DFEM) is introduced to capture fine-grained details and dynamically adjust to differences between extracted features. A ResNet-based multibranch architecture, combined with a CMFM module, integrates these features into a unified set, which improves robustness and generalizability across diverse fire detection scenarios.

The proposed method is tested on images collected from the internet and in the real world. The experimental results show that our proposed method significantly outperforms existing advanced models under various environmental conditions. The feasibility and superior performance of the algorithm in fire detection are demonstrated.

The remainder of this paper is organized as follows. Section 2 introduces the method and theory. Section 3 describes the dataset in detail, including the production of training, validation, and test sets. Section 4 describes the implementation, including aspects of the neural network setup, robotic detection trials, and evaluation criteria, among others. Insights into the experimental results and their subsequent analysis are presented in Section 5. Section 6 presents the conclusions.

## 2. Methodology

The overall architecture is illustrated in Fig. 1. The system begins with the YOLOv8 model for initial object detection and segmentation, which identifies potential fire regions in the input image. The ability

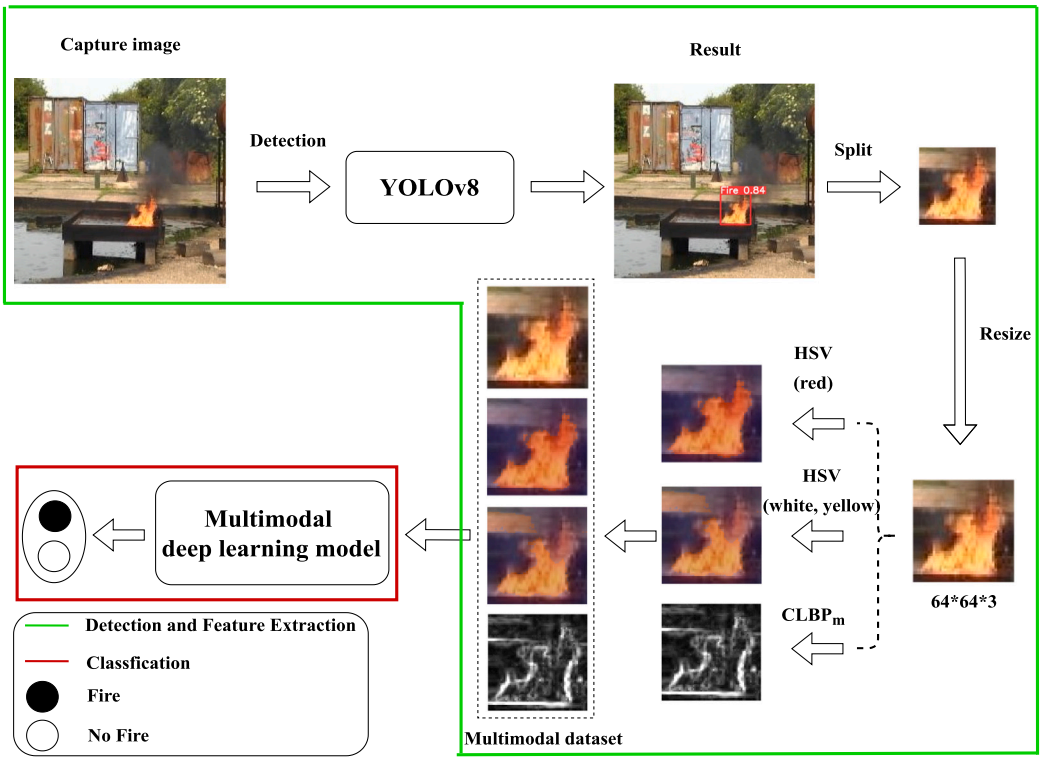



**Fig. 1.** Overview of the modified framework.

of YOLOv8 to detect objects in real time ensures that no fire-related areas are overlooked. These regions are then processed, including via hue, saturation, value (HSV) color enhancement and completed local binary pattern (CLBP) texture analysis, by feature extraction modules to capture both color and texture cues essential for fire detection. The deep feature enhancement module (DFEM) further refines the texture features and enhances subtle differences that are critical for identifying fires, even under challenging conditions. These enhanced features are passed through separate ResNet branches to process each modality individually, which ensures that each type of information is handled with maximum efficiency. Finally, the cross-modal fusion module (CMFM) merges these features to form a comprehensive representation, which is then used for classification. The synergistic combination of these techniques improves the system's robustness, thus making it highly reliable in diverse fire detection scenarios.

### 2.1. Object detection and segmentation with YOLOv8

To concentrate on potential fire regions, we employ the YOLOv8 model on a camera-captured image $I_{\text{camera}}$. YOLOv8 uses a neural network optimized for real-time object detection and is configured with a lower detection threshold $\tau_{\text{YOLO}}$ to ensure comprehensive coverage of potential fire areas. While lowering the threshold may result in false positives, it prioritizes recall to ensure that no fire-related areas are missed. This strategy is beneficial when safety is paramount and when even a subtle indication of fire is unacceptable.

The YOLOv8 detection process can be formulated as follows:

$$B = \text{YOLOv8}(I_{\text{camera}}; \theta_{\text{YOLO}}) \tag{1}$$

where $B = \{b_i\}$ is the set of detected bounding boxes and where $\theta_{\text{YOLO}}$ represents the model parameters. Each bounding box $b_i$ is described by coordinates $(x_i, y_i)$ and dimensions $(w_i, h_i)$.

For each detected box $b_i$, we extract a region:

$$I_i = I_{\text{camera}}[x_i : x_i + w_i,\ y_i : y_i + h_i] \tag{2}$$

The resulting set of regions $\{I_i\}$ serves as input for subsequent feature extraction and analysis, which ensures that computational resources are concentrated on regions likely to contain fires.

### 2.2. Feature extraction: Integration of color and texture

After applying YOLOv8 for initial object detection and segmentation, we focus on the potential fire regions identified by the model. These regions, which may contain flames in varying stages and conditions, need to be analyzed to distinguish features. To effectively represent the presence and characteristics of flames, we extract complementary features from both color and texture. Color provides valuable information about the flame temperature and combustion stages, whereas texture captures structural and dynamic patterns that are not always visible through color alone. By combining these two modalities, we aim to increase the robustness of fire detection under diverse conditions. This complementarity guides our choice to combine HSV color enhancements with CLBP texture features.

#### 2.2.1. Original image

The raw region $I_i$ retains all unmodified information. The inclusion of this original representation ensures that our model can fall back on native pixel intensities and patterns, which may contain subtle cues overlooked by processed modalities.

#### 2.2.2. HSV color feature extraction

We convert $I_i$ into the HSV space:

$$I_{\text{HSV},i} = \text{HSVConversion}(I_i) \tag{3}$$

The HSV representation decouples color from brightness, making it more robust to lighting changes and more aligned with human color perception. To highlight flames at different combustion stages, we define the following color ranges:

$V_{\text{red}}$ captures red flames, which are typically lower-temperature or early-stage flames. The hue $H$ is selected within the red portion of the

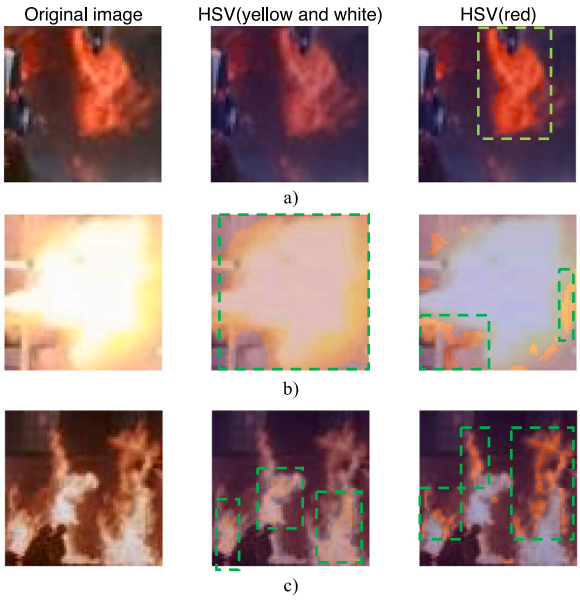



**Fig. 2.** Enhanced representation of flames in the HSV color space by different colors. (a) Red flames; (b) White and yellow flames; (c) Red, white, and yellow flames. For each column, from left to right, the images are displayed in the following order: original image, HSV image emphasizing white and yellow, and HSV image emphasizing red. The green dotted lines mark the highlighted features.

spectrum, and the saturation $S$ and value $V$ are constrained to ensure that the flame has sufficient color intensity and brightness, respectively.

$$V_{\text{wy}} = \{x \mid x(H) \in [15°, 35°], x(S) \in [0, 255], x(V) \in [200, 255]\} \tag{4}$$

$V_{\text{wy}}$ captures white-yellow flames, which are characteristically higher-temperature flames. The hue $H$ is chosen to cover yellowish tones, whereas the saturation and value ranges focus on bright, intense flames.

$$V_{\text{red}} = \{x \mid x(H) \in [0°, 10°] \cup [160°, 180°], \\ x(S) \in [100, 255], x(V) \in [100, 255]\} \tag{5}$$

where $x(H), x(S), x(V)$ are the hue, saturation, and value, respectively, of pixel $x$. These ranges are designed to highlight the key color characteristics of flames at different combustion temperatures. Then, we define masks to isolate the relevant color regions in the HSV space on the basis of the predefined ranges for red and white-yellow flames:

$$M_{\text{red},i}(x) = \begin{cases} 1 & \text{if } x \in V_{\text{red}} \\ 0 & \text{otherwise,} \end{cases} \quad M_{\text{wy},i}(x) = \begin{cases} 1 & \text{if } x \in V_{\text{wy}} \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

These masks $M_{\text{red},i}$ and $M_{\text{wy},i}$ are used to filter out pixels that fall outside the target flame color ranges. Applying these masks to the HSV representation of the flame region yields

$$I_{\text{HSV-red},i} = I_{\text{HSV},i} \odot M_{\text{red},i}, \quad I_{\text{HSV-wy},i} = I_{\text{HSV},i} \odot M_{\text{wy},i} \tag{7}$$

This process enhances the relevant flame color features by zeroing out nonflame pixels, which allows for a more accurate extraction of color information that corresponds to red and white-yellow flames. Fig. 2 illustrates the enhanced representation of different colored flames in the HSV color space. As depicted, the enhancement process distinctly accentuates the red, white, and yellow components of the flames, which makes them more definitive and pronounced in the imagery.

### 2.2.3. CLBP texture feature extraction

After enhancing the color features through HSV transformation, we turn to texture analysis, which is critical for capturing the structural and dynamic properties of flames that color alone may not fully represent. Specifically, we use the CLBP_M operator to extract texture features from $I_i$:

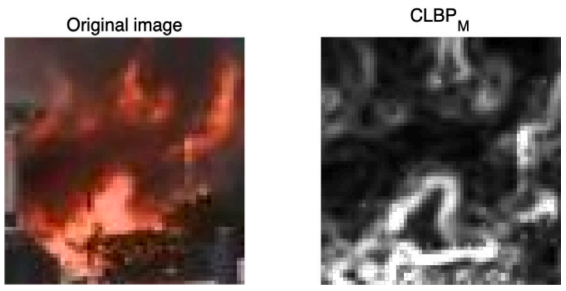



**Fig. 3.** Original image (left) and its CLBP_M image (right).

$$I_{\text{CLBP},i} = \text{CLBP}(I_i) \tag{8}$$

For a pixel $t_c(x, y)$ and its $P$ neighbors $\{t_p(x, y)\}$ within radius $R$,

$$\text{CLBP}^{\text{M}}_{P,R}(x, y) = \sum_{p=0}^{P-1} s(|t_p - t_c| - c)2^p \tag{9}$$

with

$$s(u) = \begin{cases} 1 & u \geq 0 \\ 0 & u < 0, \end{cases} \quad c = \frac{1}{N_i} \sum_{(x,y)} \sum_{p=0}^{P-1} |t_p(x, y) - t_c(x, y)| \tag{10}$$

where $N_i$ is the total number of pixels in $I_i$. CLBPs capture the magnitudes of local intensity differences, thus providing a robust measure of texture that helps distinguish fire patterns from nonfire backgrounds. Fig. 3 shows an example of the CLBP_M of an input segmented image. The texture features of the flames are distinctly observable in the image.

## 2.3. Multimodal input network structure

To make full use of the complementary characteristics of multimodal input, this method uses a multimodal input network structure. The network receives feature inputs from different modalities, including the original image, the HSV-enhanced image (red and white-yellow features), and the image obtained by enhancing the texture features through a DFEM. For each modality, feature extraction is performed through an independent ResNet branch to ensure that the information of each modality is fully mined without interference. In the feature fusion stage, a CMFM module is used to integrate these modal features, and the key cross-modal information is extracted through efficient channel attention (ECA) and efficient 2D-selective scan module (E2DSM) mechanisms. In addition, a DFEM further enhances the texture details, thereby improving the ability to distinguish fire areas. This design effectively combines the advantages of color, texture, and original features to provide a robust representation capability for fire detection. The multimodal model structure is shown in Fig. 4. The specific structure of each module is introduced below.

### 2.3.1. Dynamic feature enhancement module (DFEM)

In complex vision tasks such as fire detection, subtle texture cues are often crucial for accurate identification. Even though CLBPs can highlight certain local patterns, they may still fail to fully emphasize fine-grained texture information, or they might inadvertently amplify noise. We introduce the DFEM to refine texture features adaptively and focus dynamically on salient differences between the original and CLBP-based representations.

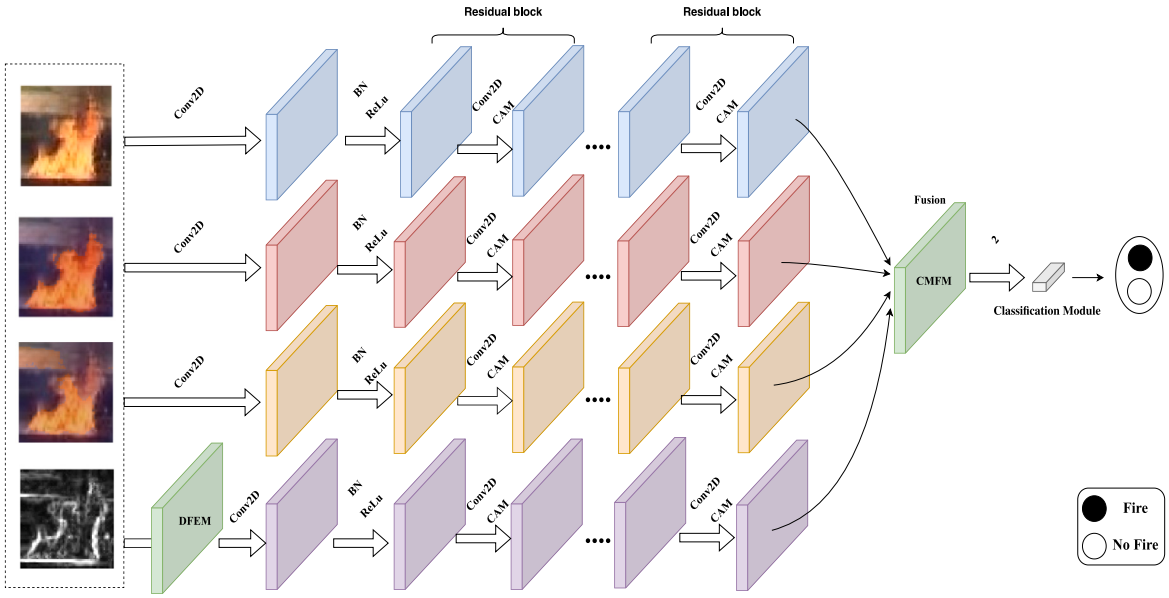



**Fig. 4.** Multimodal model structure.

a. Difference Feature Calculation

To better exploit the complementary information between the raw image and its CLBP counterpart, we first compute the difference feature. Given an input image $I_i$ and its CLBP-transformed version $I_{\text{CLBP},i}$,

$$\Delta I_i = I_i - I_{\text{CLBP},i} \tag{11}$$

This difference $\Delta I_i$ isolates subtle or complementary textural details that may not be adequately captured by either $I_i$ or $I_{\text{CLBP},i}$ alone. In other words, $\Delta I_i$ reveals how the CLBP emphasizes certain texture patterns differently than the raw image does, which provides a roadmap for selective texture refinement.

b. Learnable Descriptive Convolution (LDC)

Although $\Delta I_i$ provides an important cue regarding texture discrepancies, the model also needs a way to refine and enhance fine-grained structures within the original input $I_i$. To achieve this, we apply LDC:

$$L_i = \text{LDC}(I_i) \tag{12}$$

LDC is inspired by the dynamic feature extraction techniques used in FusionMamba's modules [19]. It utilizes a set of learnable filters whose weights are trained to respond adaptively to local patterns. By focusing on salient textures (e.g., edges, corners, and intricate flame patterns), LDC helps emphasize fine details that are critical for accurate fire region discrimination. The structure of LDC is shown in Fig. 4.

c. Dynamic Difference Perception Attention (DDPA)

Having extracted both the original refined texture representation $L_i$ and the difference information $\Delta I_i$, the next step is to determine which regions of the image warrant heightened attention. Not all texture differences are equally informative—some differences may stem from irrelevant noise or noncritical details. To address this, we introduce a dynamic difference perception attention (DDPA) module. The DDPA assigns a learned, spatially varying importance weight based on $\Delta I_i$:

$$W_i = \sigma(\text{Conv}(\Delta I_i)) \tag{13}$$

where $\sigma(\cdot)$ is the sigmoid function and $\text{Conv}(\cdot)$ learns attention weights. This step adaptively emphasizes regions where texture differences are most relevant for detecting fires.

d. Feature Enhancement

The final step combines the attention-weighted enhanced textures with the original image. By applying the learned attention weights $W_i$ to the LDC output $L_i$ and then integrating the result with $I_i$, we form the final enhanced feature representation:

$$D_{\text{DFEM},i} = W_i \odot L_i + I_i \tag{14}$$

where $\odot$ denotes elementwise multiplication. $D_{\text{DFEM},i}$ is an enriched representation that merges original intensity patterns with enhanced texture cues.

The resulting output $D_{\text{DFEM},i}$ is thus a carefully balanced representation. The DFEM leverages dynamic texture enhancement concepts while selectively highlighting discriminative differences uncovered by CLBPs. In doing so, the DFEM provides a richer, more discriminative feature map, thereby enabling more accurate and robust fire detection on the basis of subtle textural cues.

### 2.3.2. ResNets for multimodal feature processing

To effectively process and integrate the diverse feature representations extracted from the original image, HSV-enhanced images, and DFEM output, we leverage ResNets. Each modality-specific representation is passed through a dedicated ResNet branch, which ensures that relevant features are captured independently before the final fusion. We now have four different feature representations for region $I_i$:

1. $I_i$ (original)
2. $I_{\text{HSV-red},i}$ (red-enhanced HSV)
3. $I_{\text{HSV-wy},i}$ (white-yellow-enhanced HSV)
4. $D_{\text{DFEM},i}$ (DFEM output)

Each representation captures a unique aspect of the fire feature. To process these effectively, we use four separate ResNet branches:

$$F_{\text{RGB},i} = \text{ResNet}_{\text{RGB}}(I_i) \tag{15}$$

$$F_{\text{HSV-red},i} = \text{ResNet}_{\text{HSV-red}}(I_{\text{HSV-red},i}) \tag{16}$$

$$F_{\text{HSV-wy},i} = \text{ResNet}_{\text{HSV-wy}}(I_{\text{HSV-wy},i}) \tag{17}$$

$$F_{\text{DFEM},i} = \text{ResNet}_{\text{DFEM}}(D_{\text{DFEM},i}) \tag{18}$$

ResNet architectures are chosen for their ability to learn hierarchical features and mitigate vanishing gradients through residual connections. Processing each modality separately ensures that the network can extract the most relevant features without intermodal interference at this stage.

### 2.3.3. Feature fusion via cmfm

After separate ResNet branches learn modality-specific features, these features must be fused to form a comprehensive representation. We propose the CMFM module to combine these outputs effectively. The CMFM module leverages efficient channel and spatial attention to

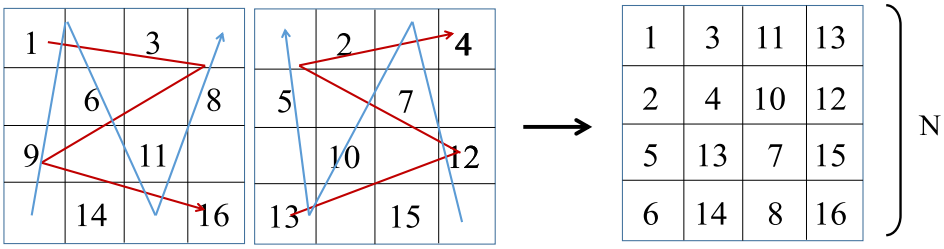



**Fig. 5.** Overall structure of E2DSM.

identify and emphasize complementary information across modalities. The overall structure is shown in Fig. 5.

a. Feature Mixing

Each ResNet branch processes a different modality. However, these modalities exist in distinct feature spaces owing to their differing focuses and processing pipelines. Feature mixing aligns these features and enables the network to leverage their combined strengths.

The process begins by transforming each modality-specific feature $F_{k,i}$ ($k \in \{\text{RGB}, \text{HSV-red}, \text{HSV-wy}, \text{DFEM}\}$) using linear transformations and depthwise convolutions. This step ensures that each feature is projected into a shared latent space where intermodality correlations can be effectively explored. Mathematically, this is expressed as follows:

$$\tilde{F}_{k,i} = \text{Dwc}(\text{Linear}(F_{k,i})) \tag{19}$$

where Linear$(\cdot)$ represents a fully connected layer that learns modality-specific weights to project each feature vector into a common-dimensional space and Dwc is the depthwise convolution operation. This step aligns the feature spaces and prepares them for fusion.

After transformation, the features are fused via elementwise multiplication ($\otimes$) and addition (+). Elementwise multiplication emphasizes shared and overlapping information among modalities, which is critical for identifying consistent fire-related patterns across color and texture features.

$$\overline{\mathbf{H}}_i = \tilde{F}_{\text{RGB},i} \otimes \tilde{F}_{\text{HSV-red},i} \otimes \tilde{F}_{\text{HSV-wy},i} \otimes \tilde{F}_{\text{DFEM},i} + \sum_k F_{k,i} \tag{20}$$

The elementwise multiplication ($\otimes$) and addition operations preserve critical information from all the modalities while highlighting the shared patterns.

b. Efficient Spatial Scanning and Channel Attention

We pass $\overline{\mathbf{H}}_i$ through the efficient 2D-selective scan module (E2DSM) to enhance spatial relationships, followed by the efficient channel attention (ECA) mechanism [20] to emphasize critical channelwise features:

$$\mathbf{H}_i = \text{LN}(\text{E2DSM}(\text{SiLU}(\overline{\mathbf{H}}_i))) \tag{21}$$

$$X_i = \text{ECA}(\mathbf{H}_i) + \sum_k F_{k,i} \tag{22}$$

The E2DSM is designed specifically to capture long-range spatial dependencies, which are essential for modeling the irregular and dynamic structures of flames. By employing a selective scanning mechanism, the E2DSM aggregates spatial information across the feature map while maintaining computational efficiency. This operation ensures that the model identifies meaningful spatial relationships, such as the transitions between fire and nonfire regions and the flickering boundaries of flames. The structure of the E2DSM is shown in Fig. 5.

Once the spatial dependencies are enriched, the ECA mechanism recalibrates the feature map adaptively along the channel dimension. By assigning attention weights to each channel on the basis of its global significance, the ECA ensures that the most informative channels—those that capture key color, texture, or intensity features—are emphasized. This dynamic adjustment complements the spatial refinement achieved by the E2DSM, which leads to a more balanced and comprehensive feature representation. The structure of the ECA is shown in Fig. 6.

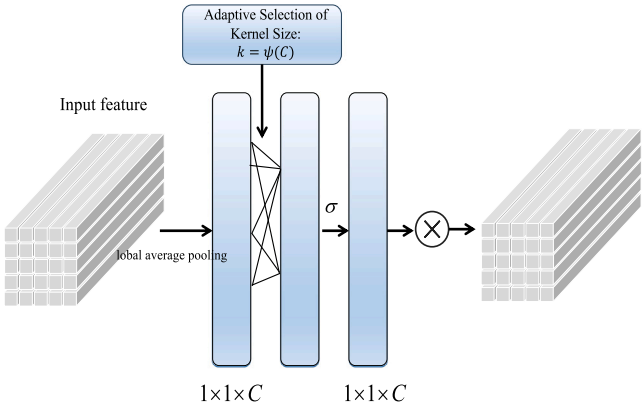



**Fig. 6.** Overall structure of ECA. ECA generates channel weights by performing fast 1D convolution on the aggregated features obtained by global average pooling, with a convolution kernel size of $k$, where $k$ is adaptively determined by mapping the channel dimension $C$.

### 2.3.4. Classification module

The fused feature $X_i$ is used for final classification by predicting the probability $\hat{y}_i$ that region $I_i$ contains fire:

$$\hat{y}_i = \text{Sigmoid}(W_{\text{clf}} X_i + b_{\text{clf}}) \tag{23}$$

where $W_{\text{clf}} \in \mathbb{R}^{1\times d}$ and $b_{\text{clf}} \in \mathbb{R}$ are the classification layer parameters and where $d$ is the dimension of $X_i$.

### 2.3.5. Loss function

To optimize the model to classify fire areas effectively, we define a classification loss function as the main objective of fire detection to ensure that the model can correctly distinguish between fire and nonfire areas. Specifically, we use the binary cross-entropy loss, which is defined as follows:

$$\mathcal{L}_{\text{cls}} = -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{24}$$

where $y_i \in \{0, 1\}$ represents the true label of region $I_i$ and $\hat{y}_i \in [0, 1]$ represents the predicted probability that the region contains a fire.

## 3. Dataset

### 3.1. Training and validation sets

To train our network, we collected 11,391 fire images that contain different scenes from the internet, as exemplified in Fig. 7. These images span a broad spectrum of environments, ranging from daylight to nighttime and from forests to urban settings, and showcase diverse flame shapes, colors, and textures. In addition, we gathered 9601 nonfire images that are not merely simple background photos but contain elements that could be confused with fire characteristics, such as sunlight and yellow and red vegetation, with various examples depicted in Fig. 8. The sample distribution is 70% for training and 30% for validation. Notably, the multimodal fusion classification

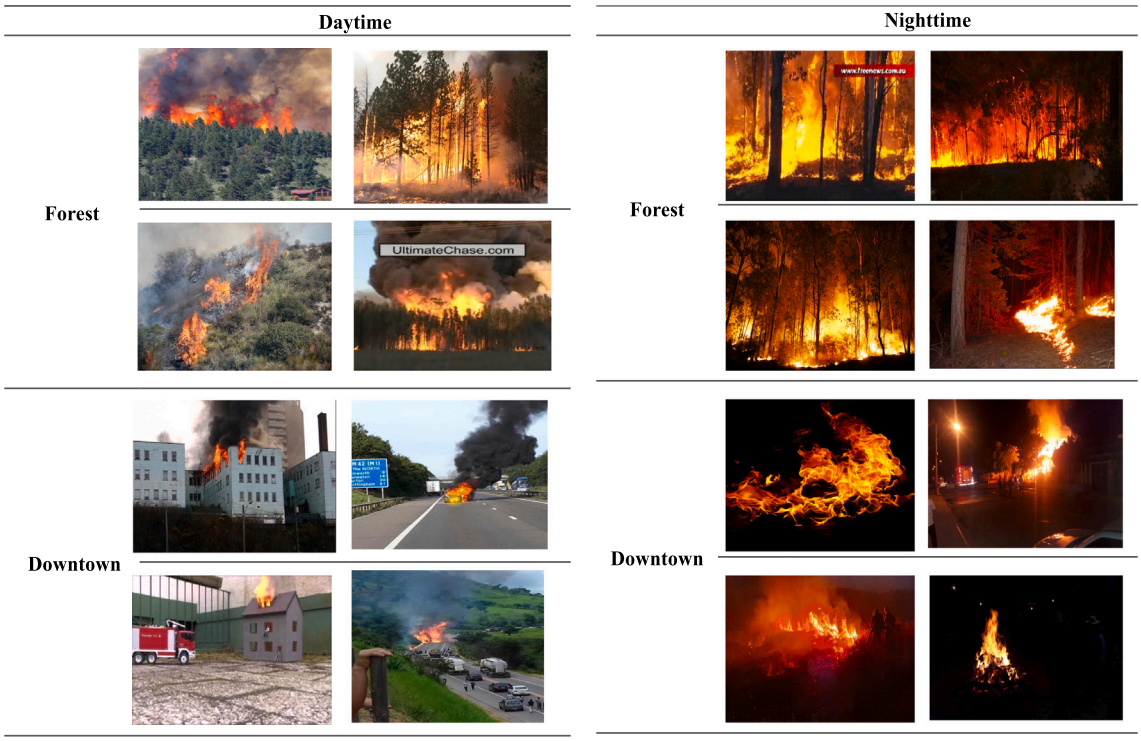



**Fig. 7.** Examples of some images from different scenes in the dataset.




**Fig. 8.** Our dataset includes four scene images, including (a) sun scene, (b) forest, (c) yellow flower, and (d) red flower.

**Table 1**
The detailed statistics of training, validation, and testing.

| Classification | Training set | Validation set | Testing set | Total |
|---|---|---|---|---|
| Fire(YOLOv8) | 7973 | 2277 | 1139 | 11 389 |
| Non Fire(YOLOv8) | 6720 | 1920 | 960 | 9600 |
| Fire(Multimodal fusion model) | 12 586 | 3596 | 1798 | 17 980 |
| Non Fire(Multimodal fusion model) | 13 194 | 3769 | 1885 | 18 848 |

model mainly processes images segmented by YOLOv8. On the basis of annotated target information, we manually segmented specific targets, such as flames and the sun, as illustrated in Fig. 9. Consequently, we obtained 17,980 fire images and 18,848 nonfire images, all of which were resized to dimensions of 64 × 64 × 3 using bicubic interpolation. Detailed statistics on the training and validation data for YOLOv8 and the multimodal fusion network are presented in Table 1.

### 3.2. Test set

During the robot navigation and image acquisition experiments, a mobile robot collected data in diverse environments, including indoors, corridors, outdoor sidewalks, shrubs, and forests. Fig. 10 displays a real-life scene of a forest site. The robot used to capture the images was equipped with four omnidirectional wheels and single-line light detection and ranging. A land robot equipped with a camera is shown in Fig. 11, and its control system consists of an NVIDIA Jetson Nano module and a personal computer that runs a robot operating system (ROS) on an STM32 microcontroller. To assess the effects of different lighting conditions on image acquisition, experiments were conducted during both the daytime and the nighttime. Throughout the experiments, the robot was remotely maneuvered manually, which enabled the camera to capture images from various angles and distances. The robot collected images at a resolution of 640 × 480 at a rate of two frames per second. A total of 847 images of fires at various times and in various scenes and 702 nonfire images of forest scenes were captured, with 1549 images in total. To ensure the safety of the experiments, simulated flames were used at different sites, thus posing no risk of actual fires. Examples of images captured by the robot camera in

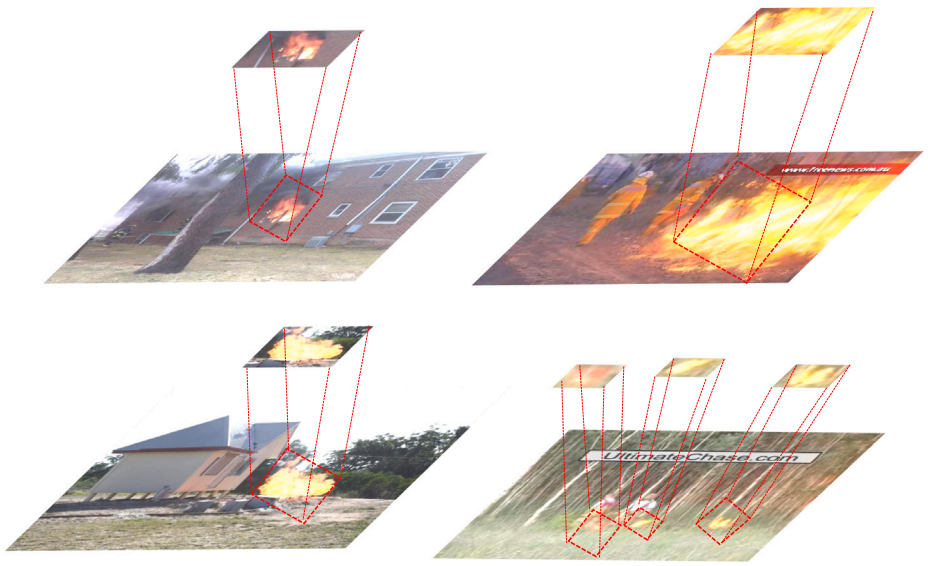


**Fig. 9.** Examples of extracting targets to be detected in different scenarios in the dataset.



**Fig. 10.** Forest experiment scene.

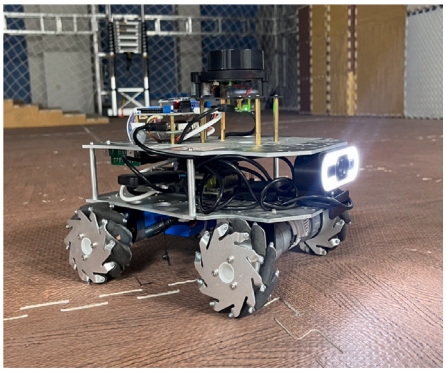


**Fig. 11.** The physical model of mobile robot.

different scenarios are shown in Fig. 12. Fig. 12 displays a variety of images captured by the robotic camera in different environments. These images reveal that daytime flames are less distinct than nighttime flames because of sunlight interference, which makes their detection more challenging. Additionally, the varied nature of backgrounds poses further complications in identifying flames.

## 4. Implementation details

Before delving into the details of the proposed model, this paper outlines the configuration of the experiment. We employed PyTorch [21] as the primary deep learning framework to implement YOLOv8 and the multimodal fusion model. The experiments were conducted on a computer that was equipped with an Intel(R) Core(TM) i7-10700F CPU @ 2.90 GHz and had 64.0 GB of RAM and a 64-bit operating system to ensure the efficiency of data processing and system stability. For the model training phase, we utilized an NVIDIA A100 Tensor Core GPU to take advantage of its powerful parallel processing capabilities to accelerate the training process. The experimental operations were based on the Ubuntu 18.04 operating system, which ensured consistency and compatibility of the environment. In this section, we describe the implementation of the proposed method in detail, including the integration process of each component and the setting of hyperparameters for model training.

In our research, we opted for the YOLOv8n model, which is a lightweight variant within the publicly available YOLOv8 series that is distinguished by fewer parameters and expedited inference speed.[1] The hyperparameter configuration for the YOLOv8n model employed in the experiments is outlined in Table 2. To normalize the input data, all the images were resized to dimensions of $640 \times 640 \times 3$ prior to being processed by the model. The role of YOLOv8 was to delineate all suspected flame areas; hence, a lower object detection confidence threshold $\tau = 0.1$ was set. Although this approach may increase false alarms, it significantly improves the thoroughness of fire detection. Given that the sizes of these segments varied, they were uniformly resized to $64 \times 64 \times 3$ through interpolation. These resized fragments were subsequently forwarded to a texture and color feature extractor for further analysis.

The hyperparameter configuration for the ResNet-based multimodal model employed in the experiments is outlined in Table 3.

### 4.1. Robot detection experiment

In this work, we developed an integrated scheme for robotic image acquisition and fire detection. The specific detection process is shown in Fig. 13. The detection network has been introduced previously.

[1] See the Ultralytics official GitHub repository: https://github.com/ultralytics/ultralytics.

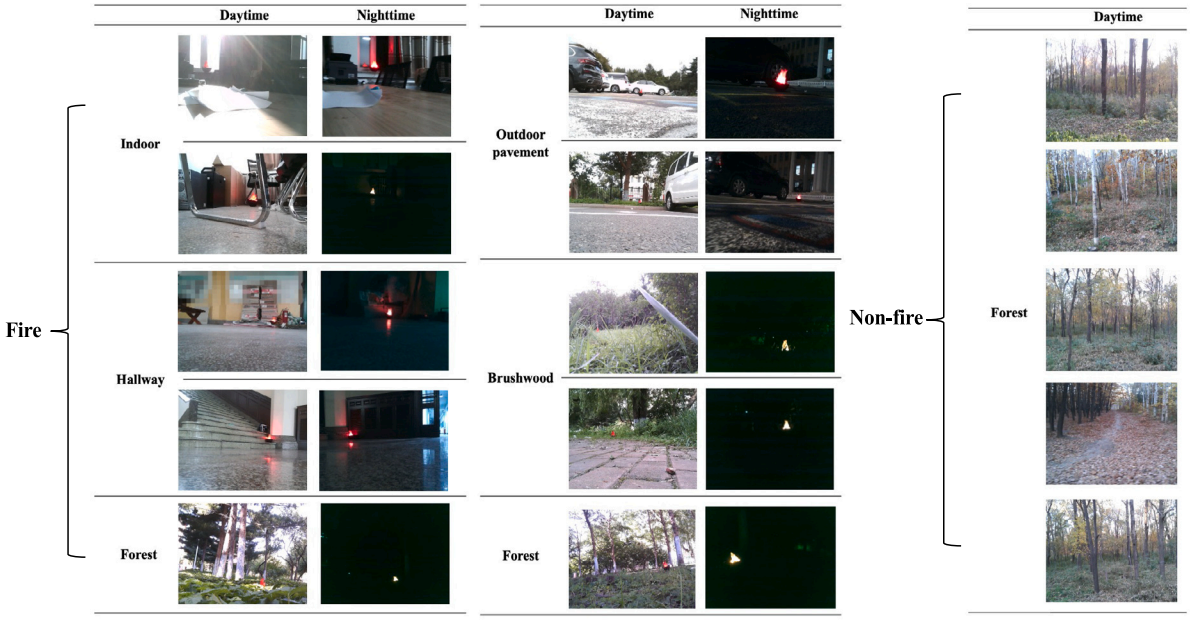



**Fig. 12.** Examples of some images captured using the camera in different scenarios.

**Table 2**
Hyperparameters for the YOLOv8.

| Hyperparameters | Value |
|---|---|
| Epochs | 150 |
| LR | 0.01 |
| Batch-size | 8 |
| Image-size | 640 |
| Optimizer | SGD |

**Table 3**
Hyperparameters and their values in the multimodal model.

| Hyperparameter | Value |
|---|---|
| Epochs | 200 |
| Batch-size | 64 |
| LR | 0.0001 |
| Dropouts | 0.3 |
| Optimizer | Adam |

Here, we introduce the process of manually controlling the robot and transmitting the captured images to the server for analysis.

(1) Robot motion control

After determining the architecture of the image transmission, the next task was to control the robot's continuous motion to capture images in a complex and dynamic environment, where a human controlled the robot remotely. The robot was first initialized using the "roscore" command in ROS2 to activate the desired environment. After that, the "ros2 run teleop_twist_keyboard teleop_twist_keyboard" command was executed to ensure that the host computer could remotely control the robot to move and operate accordingly.

(2) Image transmission system

Owing to network bandwidth constraints, we chose to capture still images periodically to minimize the significant delay that can result from high-frequency transmission. Since the robot performed image acquisition in an environment far from the server, establishing a connection with the server under the same LAN was not possible, and image transfer was dependent on the internet. During the robot acquisition process, two images were captured every second.

**Table 4**
Confusion matrix in the case of two classifications.

| | | Fire | Non-fire |
|---|---|---|---|
| True label | Fire | TP | FN |
| | Non-fire | FP | TN |

### 4.2. Assessment metrics

Table 4 presents the binary classification confusion matrix adopted in this paper. The number of true positives (TP) is the number of instances where the positive class is correctly identified, whereas the number of true negatives (TN) is the number of instances where the negative class is accurately recognized. Conversely, the number of false negatives (FN) and the number of false positives (FP) refer to the numbers of instances where the positive and negative classes, respectively, are incorrectly classified. The metrics for assessing the efficacy of the model are calculated on the basis of these values.

(1) Precision is the fraction of true positive instances among the instances labeled positive.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{25}$$

(2) Recall quantifies the fraction of actual positive instances that are correctly identified.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{26}$$

(3) The F score is the harmonic mean of precision and recall.

$$\text{F} - \text{score} = 2\left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}\right) \tag{27}$$

(4) Accuracy is the proportion of true results (both true positives and true negatives) in the complete dataset.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}} \tag{28}$$

## 5. Results and discussion

### 5.1. Network training

Following the construction of the dataset and initialization of the network, the network was trained. The training set was excluded entirely from testing and was divided into a training set and a validation

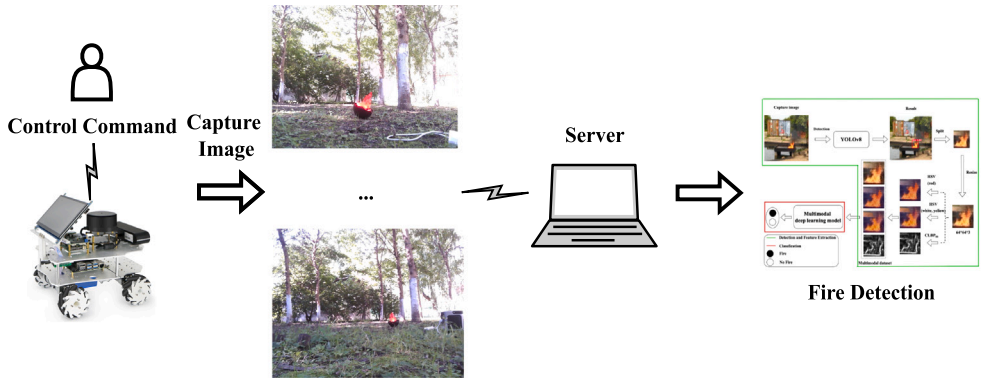



**Fig. 13.** The specific process of detecting fires using robots and cameras. By controlling the robot to capture the image, the image is uploaded to the server. Subsequently, the input image is detected using the proposed method.

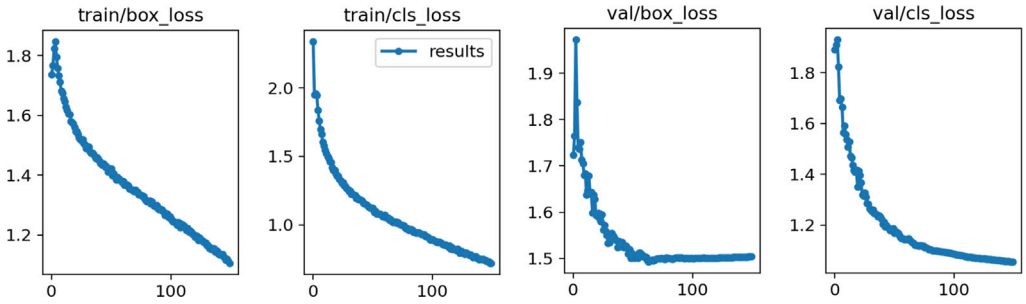



**Fig. 14.** The loss curves for the training and validation sets during the training process of YOLOv8.

set at a ratio of 70% to 30%. The training outcomes for YOLOv8 and the multimodal fusion model are presented below.

(a) YOLOv8 training

Fig. 14 shows the loss curves of YOLOv8 during training. The bounding box loss (box_loss) represents the discrepancy between the predicted and actual bounding boxes, which reflects the model's accuracy in locating the target boundaries. The class loss (cls_loss) focuses on the accuracy of classification and indicates the model's performance in recognizing and correctly categorizing detected objects. As evident from Fig. 14, both metrics declined rapidly in the early stages of training, demonstrating the model's effective learning and adaptation to the data. Correspondingly, the validation set loss also exhibited a steady downward trend. After approximately 100 training iterations, the box _loss began to stabilize. To mitigate the risk of overfitting, training was terminated at this point, and the model was saved.

(b) Multimodal fusion model training

Fig. 15 shows the loss curves of the multimodal fusion classification model during training. Upon observing that the loss curves had plateaued, which indicated that the model had adequately learned from the data, training was concluded, and the model was saved.

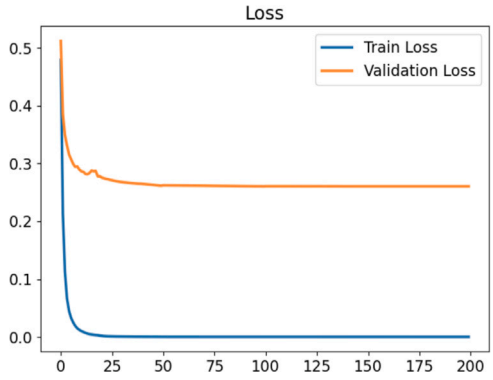



**Fig. 15.** The loss curves for the training and validation sets during the training process of multimodal network.

### 5.2. Experimental results and discussion

To demonstrate the superiority of the proposed method, this section details comparative experiments that were conducted on the test dataset. The performance of our approach was juxtaposed with those of the current state-of-the-art detection and classification networks. The models for comparison included VGG16, ResNet50, GoogLeNet, InceptionV3, SqueezeNet, MobileNet, EfficientNetB0, Vision Transformer (ViT), YOLOv5, YOLOv8, VGG16-Fine Tuner [22], InceptionV3-Fine Tuner [22], Xception-Fine Tuner [22] , and DEYO [23], among others. The training and testing sets for all the models were consistent with those used for the proposed method.

#### 5.2.1. Comparison with state-of-the-art methods

According to the results in Table 5, the classification performances of different detection methods, including traditional models and advanced deep learning architectures, in the fire detection task can be comprehensively evaluated. Compared with all the other models listed in the table, the proposed method performed well in terms of all the evaluation indicators and achieved the best performance level. Specifically, the proposed method achieved a precision of 96.4%, which was significantly higher than those of the other models, among which the second highest was 93.2% for the DEYO model. This high precision indicates that the method has a lower false alarm rate when identifying real fire events, thus ensuring the reliability and accuracy of detection. Compared with YOLOv8, the proposed methodology resulted in a slightly lower recall, which was a consequence of its cascaded structure, where YOLOv8 serves as the preliminary phase in fire detection, followed by a multimodal fusion model for further classification refinement. While such an arrangement may lead to an accumulation

**Table 5**
In the simulation experiment, the proposed method is compared with different detection methods.

| Model | Precision | Recall | F-score | Accuracy |
|---|---|---|---|---|
| Vgg16 [24] | 88.6 | 65.3 | 74.5 | 75.1 |
| Resnet50 [25] | 93.7 | 70.1 | 80.3 | 85.7 |
| Googlenet [26] | 92.3 | 57.1 | 71.4 | 80.5 |
| InceptionV3 [27] | 80.7 | 60.5 | 69.2 | 68.8 |
| SqueezeNet [28] | 84.2 | 44 | 58.1 | 73.8 |
| MobileNet [29] | 86.7 | 73.2 | 79.4 | 78 |
| EfficientnetB0 [30] | 86.6 | 68.4 | 76.4 | 75.6 |
| YOLOv5 | 84.9 | 92.9 | 88.7 | 86.5 |
| Vision Transformer(ViT) [17] | 86 | 49.6 | 63.2 | 75.9 |
| YOLOv8 | 85.5 | **95.9** | 90.3 | 88.8 |
| VGG16–Fine Tuner [22] | 89.5 | 74.9 | 81.5 | 81.3 |
| InceptionV3–Fine Tuner [22] | 88.2 | 72.9 | 79.6 | 82.8 |
| Xception–Fine Tuner [22] | 89.1 | 84.6 | 87.2 | 88.6 |
| DEYO [23] | 93.2 | 94.7 | 94.7 | 94.4 |
| Proposed method | **96.4** | 95.7 | **96** | **95.8** |

of errors, thereby diminishing recall, this slight reduction in recall by 0.5% was offset substantially by the significant improvements in other performance metrics; precision was enhanced by 10.9%, the F-score was increased by 5.7%, and accuracy was increased by 7%. In terms of the F-score, the proposed method achieved a value of 96%, which was significantly better than those of all the other models, including 94.7% for DEYO. This metric combines precision and recall and reflects the method's excellent ability to balance the two, thus indicating that it has excellent overall performance in fire detection tasks. In addition, the proposed method also achieved 95.8% accuracy, thereby outperforming all the comparison models. This result further verifies the overall superiority of the method in terms of different performance indicators and demonstrates its high efficiency and reliability in practical applications. In summary, the proposed multimodal fusion model achieves excellent performance in fire detection tasks by combining color and texture features and utilizing advanced feature enhancement and attention mechanisms. These results show that the method has greater reliability and effectiveness in practical applications, can significantly improve the overall capabilities of fire detection systems, and can ensure accurate and efficient identification of fires in various complex environments to ensure safety.

#### 5.2.2. Comprehensive performance evaluation via statistical methods

To further evaluate and compare the performance of different models or algorithms on the same dataset, we used the Friedman and Nemenyi combined analysis to provide statistical proof for selecting the best model for fire detection. Four data subsets from the test set were randomly selected to test all the comparison networks in Table 5. The number of samples in each data subset was 200. Finally, the recall, precision, F-score and accuracy of each model were obtained. We chose the F-score as the main performance evaluation criterion. First, the Friedman test was performed on the F-scores of these algorithms. The results are shown in Table 6. As shown in the table, the proposed method performed the best on multiple datasets. This method not only achieved excellent results on a single dataset but also maintained a stable high ranking in the comprehensive evaluation across datasets, especially compared with other algorithms such as DEYO and Xception–Fine Tuner. There were significant differences between different algorithms (the p value was much less than 0.05), which prompted us to conduct a more in-depth analysis and use Nemenyi analysis for subsequent tests. The results are shown in Table 7. As shown in the table, the p values between the proposed method and other methods (especially VGG16, ResNet50 and ViT) were very low, which means that the performance of the proposed method was significantly better than those of these methods. This further demonstrates the effectiveness and robustness of the proposed method in fire detection tasks.

#### 5.2.3. Confusion matrices of the proposed method

Table 8 provides a detailed breakdown of the diagnostic outcomes for the proposed fire detection method and its component—the YOLOv8 model. Fig. 16 is a visualization of Table 8, and the results are presented comprehensively. In the experiment, we analyzed the confusion matrices of YOLOv8 and the proposed fire detection method. The confusion matrix of the YOLOv8 model shows that the model correctly identified 812 positive samples (TP), misclassified 35 as negative samples (FN), correctly classified 564 negative samples (TN), and misclassified 138 negative samples as positive samples (FP). In comparison, the confusion matrix of the proposed method showed slightly different results. The method correctly identified 811 positive samples (TP), missed 36 positive samples (FN), and showed significant improvements in negative sample classification by correctly classifying 672 negative samples (TN) and misclassifying only 30 positive samples (FP). In comparison, the proposed method performed significantly better than YOLOv8 in reducing false positives (FP). This shows that the proposed multimodal feature fusion method has the potential to improve overall detection performance and reduce false positives, especially in fire areas in complex environments.

### 5.3. Ablation experiments

To comprehensively evaluate the effectiveness and contribution of each module in the proposed method, a series of ablation experiments were conducted. The main purpose of the ablation experiments was to analyze the effects of key components on the overall fire detection performance by gradually removing or replacing them to verify the importance of each module in the model. Specifically, we designed the following experimental settings:

Structure 1: Only HSV and CLBP feature extraction were retained, and no enhancement modules were added.

Structure 2: The DFEM was removed, and the other modules were retained, but dynamic feature enhancement was not performed.

Structure 3: The CMFM was removed, the other modules were retained, and only the independent features of each modality were used for classification.

Structure 4: The E2DSM was removed, the other modules were retained, and the attention mechanism was not applied in the fusion stage to analyze the effects of the attention mechanism on feature selection and weight allocation.

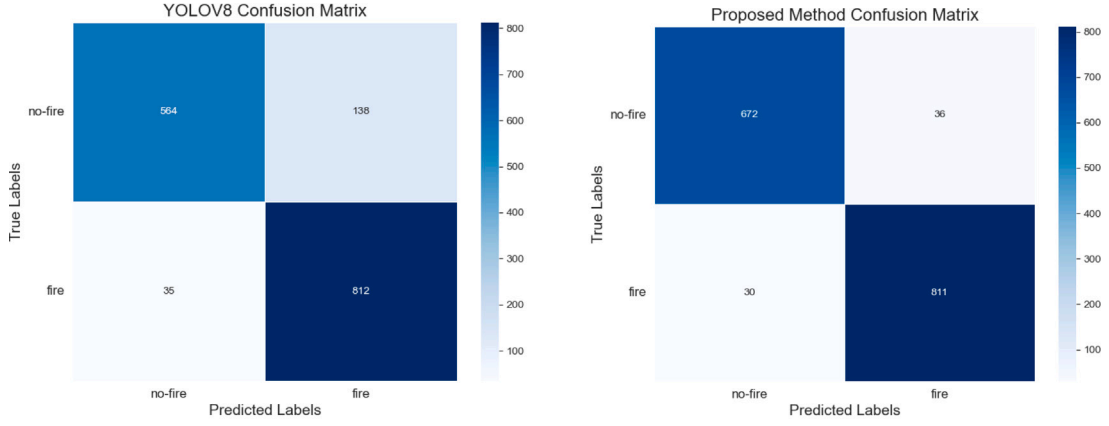
In each experimental setting, we used the same training and test datasets and used evaluation indicators such as precision, recall, F-score, and accuracy to evaluate performance. By comparing the performance differences between each ablation model and the complete model in terms of various indicators, we quantified the specific contribution of each module to the overall fire detection performance. The experimental results are shown in Table 9. The ablation experimental results in Table 9 show that each module of the proposed method contributes significantly to the overall fire detection performance. Structure 1 uses only HSV and CLBP feature extraction, and its performance was significantly lower than that of the complete model, thus indicating that the enhancement module plays a key role in improving detection precision and accuracy. After removing the dynamic feature enhancement module (Structure 2), although the performance improved, it was still lower than that of the complete model, which indicates the importance of the DFEM in refining texture features. The removal of the cross-modal fusion module and the attention mechanism in Structures 3 and 4, respectively, led to slight decreases in performance. The results emphasize the role of the CMFM and E2DSM in feature integration and selection. Overall, the complete multimodal fusion model achieved the highest precision and competitive recall and F score, which verifies the effectiveness of the synergy of each module in improving the accuracy and robustness of the fire detection system.

**Table 6**
Use friedman to test the ranking of different algorithms on each data set.

| Dataset | VGG16 | ResNet50 | GoogLeNet | InceptionV3 | SqueezeNet | MobileNet | EfficNetB0 | YOLOv5 | ViT | YOLOv8 | VGG16 Fine Tuner | InceptionV3 Fine Tuner | Xcept Fine Tuner | DEYO | Proposed Method |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Dataset1 | 11 | 7 | 12 | 13 | 15 | 9 | 10 | 4 | 14 | 3 | 6 | 8 | 5 | 2 | 1 |
| Dataset2 | 11 | 7 | 12 | 13 | 15 | 9 | 10 | 4 | 14 | 3 | 6 | 8 | 5 | 2 | 1 |
| Dataset3 | 11 | 8 | 12 | 13 | 15 | 9 | 10 | 4 | 14 | 3 | 6 | 7 | 5 | 2 | 1 |
| Dataset4 | 11 | 9 | 12 | 13 | 15 | 8 | 10 | 4 | 14 | 3 | 6 | 7 | 5 | 2 | 1 |

**Table 7**
P value for Nemenyi test.

| Dataset | VGG16 | ResNet50 | GoogLeNet | InceptionV3 | SqueezeNet | MobileNet | EfficNetB0 | YOLOv5 | ViT | YOLOv8 | VGG16 Fine Tuner | InceptionV3 Fine Tuner | Xcept Fine Tuner | DEYO | Proposed Method |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| VGG16 | 1 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.6327 | 0.9 | 0.4233 | 0.9 | 0.9 | 0.8362 | 0.2226 | 0.0991 |
| ResNet50 | 0.9 | 1 | 0.9 | 0.9 | 0.5819 | 0.9 | 0.9 | 0.9 | 0.7854 | 0.9 | 0.9 | 0.9 | 0.9 | 0.8871 | 0.6836 |
| GoogLeNet | 0.9 | 0.9 | 1 | 0.9 | 0.9 | 0.9 | 0.9 | 0.4233 | 0.9 | 0.2226 | 0.8362 | 0.9 | 0.6327 | 0.0991 | 0.0378 |
| InceptionV3 | 0.9 | 0.9 | 0.9 | 1 | 0.9 | 0.9 | 0.9 | 0.2226 | 0.9 | 0.0991 | 0.6327 | 0.9 | 0.4233 | 0.0378 | 0.0124 |
| SqueezeNet | 0.9 | 0.5819 | 0.9 | 0.9 | 1 | 0.7854 | 0.9 | 0.0378 | 0.9 | 0.0124 | 0.2226 | 0.5309 | 0.0991 | 0.0036 | 0.001 |
| MobileNet | 0.9 | 0.9 | 0.9 | 0.9 | 0.7854 | 1 | 0.9 | 0.9 | 0.9 | 0.8871 | 0.9 | 0.9 | 0.9 | 0.6836 | 0.4794 |
| EfficientNetB0 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 1 | 0.8362 | 0.9 | 0.6327 | 0.9 | 0.9 | 0.9 | 0.4233 | 0.2226 |
| YOLOv5 | 0.6327 | 0.9 | 0.4233 | 0.2226 | 0.0378 | 0.9 | 0.8362 | 1 | 0.0991 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 |
| ViT | 0.9 | 0.7854 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 | 0.0991 | 1 | 0.0378 | 0.4233 | 0.7345 | 0.2226 | 0.0124 | 0.0036 |
| YOLOv8 | 0.4233 | 0.9 | 0.2226 | 0.0991 | 0.0124 | 0.8871 | 0.6327 | 0.9 | 0.0378 | 1 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 |
| VGG16–FT | 0.9 | 0.9 | 0.8362 | 0.6327 | 0.2226 | 0.9 | 0.9 | 0.9 | 0.4233 | 0.9 | 1 | 0.9 | 0.9 | 0.9 | 0.9 |
| InceptionV3–FT | 0.9 | 0.9 | 0.9 | 0.9 | 0.5309 | 0.9 | 0.9 | 0.7345 | 0.9 | 0.9 | 0.9 | 1 | 0.9 | 0.9 | 0.7345 |
| Xception–FT | 0.8362 | 0.9 | 0.6327 | 0.4233 | 0.0991 | 0.9 | 0.9 | 0.9 | 0.2226 | 0.9 | 0.9 | 0.9 | 1 | 0.9 | 0.9 |
| DEYO | 0.2226 | 0.8871 | 0.0991 | 0.0378 | 0.0036 | 0.6836 | 0.4233 | 0.9 | 0.0124 | 0.9 | 0.9 | 0.9 | 0.9 | 1 | 0.9 |
| Proposed method | 0.0991 | 0.6836 | 0.0378 | 0.0124 | 0.001 | 0.4794 | 0.2226 | 0.9 | 0.0036 | 0.9 | 0.9 | 0.7345 | 0.9 | 0.9 | 1 |




(a) (b)

**Fig. 16.** Confusion matrix obtained using (a) YOLOv8, (b) proposed method in experiments.

**Table 8**
Confusion matrices were obtained for YOLOv8 and overall framework in experiment.

| MODEL | TP | FN | TN | FP |
|---|---|---|---|---|
| YOLOv8 | 812 | 35 | 564 | 138 |
| Proposed method | 811 | 36 | 672 | 30 |

**Table 9**
Ablation experiment results.

| Model | Precision | Recall | F-score | Accuracy |
|---|---|---|---|---|
| Structure 1 | 95.5 | 95.4 | 95.5 | 95 |
| Structure 2 | 96.1 | 95.4 | 95.8 | 95.4 |
| Structure 3 | 96.2 | 96.1 | 96.2 | 95.8 |
| Structure 4 | 95.7 | 95.6 | 95.7 | 95.3 |
| Proposed method | **96.4** | **95.7** | **96** | **95.8** |

## 6. Conclusions

The proposed multimodal fire detection framework represents a significant advance in addressing the key challenge of accurate and robust fire detection under complex environmental conditions. By combining YOLOv8-based object detection, HSV color space enhancement, CLBP texture analysis, and the DFEM, the proposed framework effectively overcomes the limitations of traditional single-modal detection methods. Extensive experimental validation, including multiple datasets and robotic image acquisition in real environments, fully demonstrates the superior performance of the proposed method compared with the state-of-the-art methods.

The main contributions of this study can be summarized into three aspects. First, the adopted multimodal fusion strategy significantly alleviates the inherent shortcomings of single feature extraction methods by integrating color, texture, and spatial information. Second, the proposed CMFM module innovatively combines ECA and the E2DSM to achieve dynamic and adaptive feature fusion. Finally, the experimental results show that the proposed method achieved a precision of 96.4%, a recall of 95.7%, and an F1-score of 96% in various challenging scenarios, thus fully demonstrating its excellent performance.

The fire detection framework has a highly modular and scalable architecture that can flexibly integrate and replace individual components according to technological developments. For example, although the current implementation uses YOLOv8 for object detection, the

modular design of the framework enables easy replacement with more advanced detection models in the future. Similarly, the attention mechanism and feature fusion modules can also be optimized or replaced to adapt to emerging feature extraction and fusion techniques. Additionally, we are committed to improving computational efficiency and reducing processing time in future iterations of the framework. This flexibility ensures that the framework can not only solve current fire detection challenges but also provide a solid foundation for future technological innovations, thus ensuring its long-term effectiveness and relevance in critical safety applications.

In future work, we plan to further evaluate the performance of our proposed framework and compare our framework with alternative approaches, such as training ConvNet on ImageNet followed by transfer learning on a fire/nonfire dataset. This comparison will provide further insights into the potential advantages and limitations of different fire detection techniques and will enable us to refine and optimize our framework to achieve even better results in various real-world scenarios.

## Ethics statement

Not applicable because this work does not involve the use of animal or human subjects.

## CRediT authorship contribution statement

**Zekun Zhou:** Methodology, Resources, Validation, Writing – review & editing. **Hongyang Zhao:** Writing – original draft, Methodology, Investigation, Conceptualization. **Xingdong Li:** Supervision, Investigation. **Yi Liu:** Validation, Formal analysis, Conceptualization. **Tao Jiang:** Validation, Formal analysis. **Jing Jin:** Supervision, Investigation. **Yanan Guo:** Writing – review & editing, Validation, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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